The grating monochromator

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Abstract

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1 Introduction

As quantum theory developed, energy levels of atoms were found to be quantized, and the emission spectra of atoms were found to consist of discrete lines. By measuring the wavelengths of these lines, the energy levels of atoms could be determined, and we can verify the predictions of quantum theory. In this experiment, we will use a Czerny-Turner monochromator to measure the emission spectrum of a mercury lamp, and determine the energy levels of mercury atoms. Then we will use the same monochromator to measure the emission spectrum of a hydrogen lamp, and determine the wavelengths each emission line corresponds to.

2 Theory

In the Czerny-Turner monochromator, concave mirrors are used to collimate the beam of light falling on the grating and to focus those beams diffracted from the grating, as shown in figure 1.

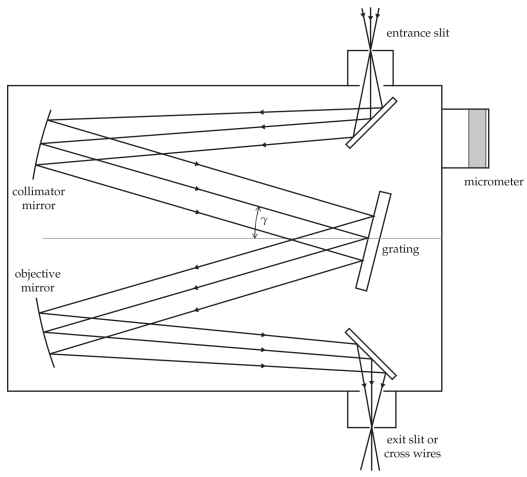

Figure 1: Czerny–Turner monochromator.

Grating size	48 mm × 48 mm
Ruling frequency ($1/d$)	1200 lines/mm
Mirror diameter	63.5 mm
Mirror radius of curvature	439 mm
Czerny–Turner angle, γ	25°28′
Length of sine-bar, l	75.24 mm

Table 1: Design parameters of the Czerny-Turner monochromator.

According to [1], the wavelength λ of the light diffracted by the grating is given by

$$\lambda = z\left(\frac{2d\cos\gamma}{pl}\right), \tag{1}$$

where z is the reading on the sine-bar, d is the grating spacing, γ is the Czerny–Turner angle, p is the ruling frequency of the grating, and l is the length of the sine-bar. Take the data in table 1, we can calculate coefficient $2d\cos\gamma/pl = 2.000 \times 10^{-5}$.

3 Experimental Setup

3.1 Calibration

The monochromator was calibrated using the known vacuum wavelengths of the mercury emission spectrum. Although the sine-bar mechanism provides an approximately linear relation between micrometer reading z and wavelength λ, mechanical imperfections and backlash in the drive mean that an empirical calibration is required.

With the mercury lamp aligned as described in figure 2, each visible spectral line was brought into coincidence with the cross-hair in the exit aperture. For each line, the micrometer was always approached from the same direction to eliminate backlash errors in the sine-bar mechanism.

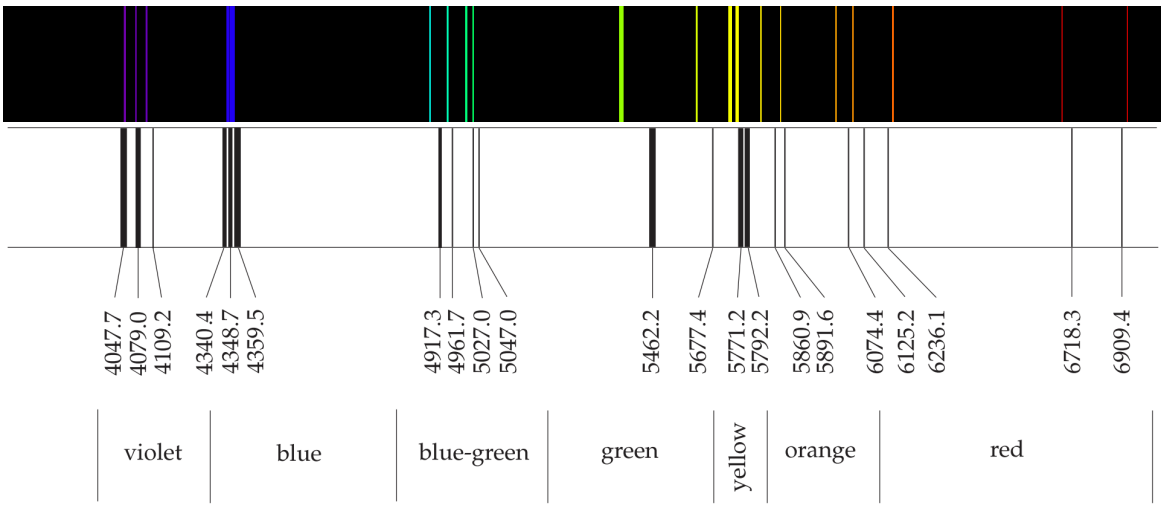

Figure 2: Corresponding mercury spectrum [1]

Three readings were taken for each spectral line and averaged. The data were recorded in table 4. The slit width was kept fixed throughout the calibration procedure and was not altered subsequently, in order to avoid systematic shifts in the apparent line centre.

The averaged micrometer readings were then paired with the corresponding known vacuum wavelengths to establish the calibration relation between z and λ.

3.2 Measurement of the Hydrogen Spectrum

After calibration, the mercury lamp was removed and replaced with the hydrogen discharge tube. The condensing lens and slit settings were left unchanged to ensure consistency with the calibration procedure.

The hydrogen source was aligned so that a uniform patch of light fell centrally on the grating. The cross-hair position was not altered after calibration, in order to preserve the established wavelength reference.

The four visible lines were identified and brought successively into coincidence with the cross-hair. As in the calibration procedure, all micrometer readings were approached from the same direction to minimise backlash effects.

For each spectral line, multiple readings were taken and averaged. The data were recorded in table 5.

4 Analysis

4.1 Calibration

We firstly performed a linear regression on the calibration data in table 4 using Python. The relationship is given by $z = a\lambda + b$, where a and b are the regression coefficients. The fitted line is shown in figure 3. The residual plot is shown in figure 4. The residuals are randomly distributed around zero, indicating that the linear model is appropriate for the calibration data.

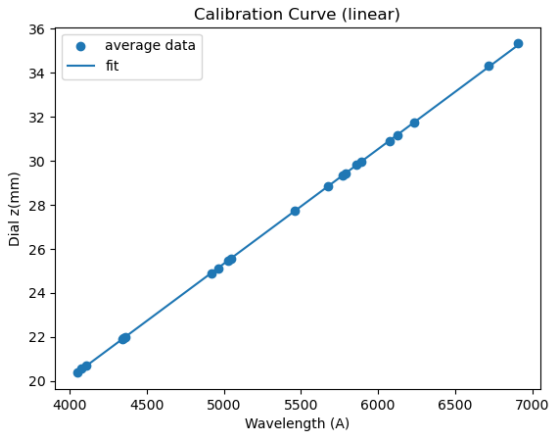

Figure 3: Calibration plot of the linear regression for calibration.

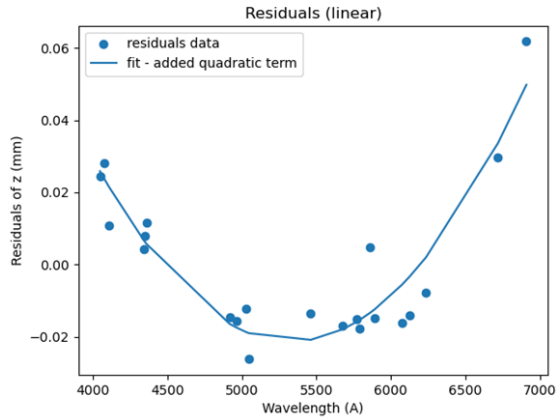

Figure 4: Residual plot of the linear regression for calibration.

It is obvious that the linear model fits the calibration data well, but the residuals show a quadratic pattern. This suggests that we should use a quadratic model to fit the calibration data. We then performed a quadratic regression on the calibration data using Python. The relationship is given by $z = a\lambda^2 + b\lambda + c$, where a, b and c are the regression coefficients. The fitted line is shown in figure 5. The residual plot is shown in figure 6. The residuals are randomly distributed around zero, indicating that the quadratic model is appropriate for the calibration data.

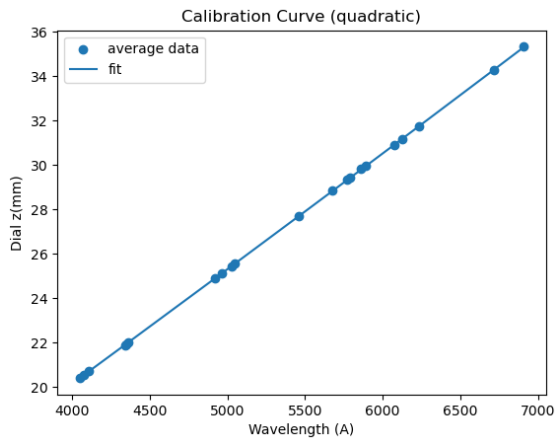

Figure 5: Calibration plot of the quadratic regression for calibration.

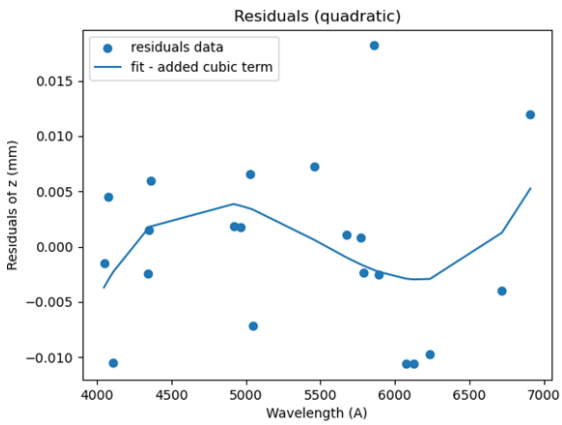

Figure 6: Residual plot of the quadratic regression for calibration.

From figure 6, we cannot see any pattern in the residuals, which suggests that the quadratic model fits the calibration data correctly. The residuals are distributed around zero, and mostly-within the range from -0.01 to 0.01, which is also the smallest resolution of the monochromator. Therefore, we can conclude that the quadratic model is appropriate for the calibration data. $R^2 = 0.999$ also indicates the goodness of fit of the quadratic model.

From the model we can obtain the three coefficients: $a = 2.856(251) \times 10^{-8}$, $b = 4.900(27) \times 10^{-5}$ and $c = 1.017(704) \times 10^{-1}$. These coefficients can be used to convert the micrometer readings of the hydrogen spectrum to wavelengths.

4.2 Hydrogen wavelengths

Using the quadratic model obtained from the calibration, we can convert the micrometer readings of the hydrogen spectrum to wavelengths. The results are shown in table 2.

Line	n	z (mm)	λ (Å)	R (10^{-3} Å^{-1})	σ_R (10^{-7} Å^{-1})
H_α	3	33.49(1)	6560(2)	1.09756	3.35
H_β	4	24.56(1)	4852(2)	1.09920	4.53
H_γ	5	21.90(1)	4338(2)	1.09772	5.06
H_δ	6	20.64(1)	4093(2)	1.09944	5.37

Table 2: Determination of the Rydberg constant from hydrogen spectra

Since we obtained σ_R for each spectral line, a weighted mean of R was calculated. The resulting value is

$$\boxed{R = 1.09828(22) \times 10^7 \text{ m}^{-1}}.$$

The uncertainty has been rounded to two significant figures, and the central value has been expressed to the corresponding precision.

4.3 Theoretical prediction

Given that $\varepsilon_0 = 8.8542 \times 10^{-12}$ F m^{-1}, $e = 1.6022 \times 10^{-19}$ C, $c = 2.9979 \times 10^8$ m s^{-1}, $\hbar = 1.0546 \times 10^{-34}$ J s, $m_e = 9.1095 \times 10^{-31}$ kg, and $m_p = 1.6726 \times 10^{-27}$ kg, we can calculate the theoretical value of the Rydberg constant using the formula

$$R_\infty = \left(\frac{1}{4\pi\varepsilon_0}\right)^2 \frac{m_e e^4}{4\pi c\hbar^3} = \boxed{10974000 \text{ m}^{-1}}.$$

The Rydberg constant for hydrogen is given by

$$R_\text{H} = \left(\frac{1}{4\pi\varepsilon_0}\right)^2 \frac{e^4}{4\pi c\hbar^3} \frac{m_e m_p}{m_e + m_p} = \boxed{10968000 \text{ m}^{-1}}.$$

4.4 Result comparisons

From sections 4.2 and 4.3, we can see that the experimental value of the Rydberg constant is in excellent agreement with theoretical predictions. However, the experimental value is slightly higher than the theoretical R_H, making it appear closer to R_∞. We suggest that this small discrepancy ($\sim 0.08\%$) is predominantly driven by systematic errors in the experiment.

Take a constant $\Delta z = 0.02$ mm error in the micrometer reading as an example. If all micrometer readings are overestimated by Δz, reconsidering the calculation of R we could find

Line	n	z (mm)	λ (Å)	R (10^{-3} Å^{-1})	σ_R (10^{-7} Å^{-1})
H_α	3	33.49(1)	6564.5 ± 2.0	1.09681	3.34
H_β	4	24.56(1)	4853.4 ± 2.0	1.09889	4.53
H_γ	5	21.90(1)	4340.4 ± 2.0	1.09711	5.05
H_δ	6	20.64(1)	4094.1 ± 2.0	1.09914	5.37

Table 3: Determination of the Rydberg constant from hydrogen Balmer lines. The wavelengths (λ) were derived from the mean micrometer readings (z) using local linear interpolation. The uncertainty in wavelength is estimated as ± 2.0 Å, which propagates to σ_R.

5 Conclusion

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References

[1] OP13: The grating monochromator

6 Appendix

Wavelength (Å)	Reading 1 (mm)	Reading 2 (mm)	Reading 3 (mm)	Average (mm)
4047.7	20.39(1)	20.39(1)	20.42(1)	20.40(1)
4079.0	20.55(1)	20.57(1)	20.58(1)	20.57(1)
4109.2	20.69(1)	20.72(1)	20.71(1)	20.71(1)
4340.4	21.90(1)	21.92(1)	21.89(1)	21.90(1)
4348.7	21.95(1)	21.95(1)	21.95(1)	21.95(1)
4359.5	22.00(1)	22.02(1)	22.01(1)	22.01(1)
4917.3	24.88(1)	24.89(1)	24.89(1)	24.89(1)
4961.7	25.12(1)	25.12(1)	25.11(1)	25.12(1)
5027.0	25.46(1)	25.46(1)	25.46(1)	25.46(1)
5047.0	25.56(1)	25.54(1)	25.55(1)	25.55(1)
5462.2	27.73(1)	27.72(1)	27.72(1)	27.73(1)
5677.4	28.84(1)	28.84(1)	28.84(1)	28.84(1)
5771.2	29.32(1)	29.33(1)	29.34(1)	29.33(1)
5792.2	29.43(1)	29.44(1)	29.44(1)	29.44(1)
5860.9	29.82(1)	29.81(1)	29.82(1)	29.82(1)
5891.6	29.96(1)	29.95(1)	29.96(1)	29.96(1)
6074.4	30.91(1)	30.89(1)	30.92(1)	30.91(1)
6125.2	31.17(1)	31.18(1)	31.17(1)	31.17(1)
6236.1	31.75(1)	31.75(1)	31.77(1)	31.76(1)
6718.3	34.29(1)	34.32(1)	34.30(1)	34.30(1)
6909.4	35.33(1)	35.33(1)	35.33(1)	35.33(1)

Table 4: Spectrometer calibration measurements (mercury lamp)

Spectral Line	Reading 1 (mm)	Reading 2 (mm)	Reading 3 (mm)	Average (mm)
H_α	33.50(1)	33.50(1)	33.48(1)	33.49(1)
H_β	24.57(1)	24.55(1)	24.56(1)	24.56(1)
H_γ	21.90(1)	21.90(1)	21.91(1)	21.90(1)
H_δ	20.66(1)	20.62(1)	20.64(1)	20.64(1)

Table 5: Spectrometer hydrogen line measurements

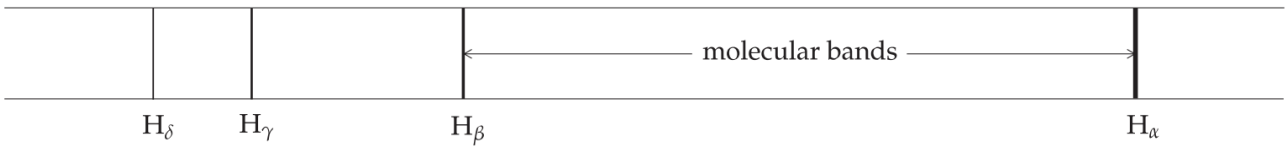

Figure 7: Corresponding hydrogen spectrum [1]